3. Critical region of a test is

(*a*) rejection region (b) acceptance region

(*c*) sample space (d) subset of the sample space

4. The critical value (table value) of the test statistic at the level of significance α for a two-tailed large sample test is

(a) $z_{\alpha/2}$ (b) z_{α}

(c) $-z_{\alpha}$ (d) $-z_{\alpha/2}$

5. In general, large sample theory is applicable when

(a) $n \geq 100$ (b) $n \geq 50$

(c) $n \geq 40$ (d) $n \geq 30$

6. When H_1 is a one-sided (right) alternative hypothesis, the critical region is determined by

(a) both right and left tails (b) neither right nor left tail

(c) right tail (d) left tail

7. Critical value at 5% level of significance for two-tailed large sample test is

(a) 1.645 (b) 2.33

(c) 2.58 (d) 1.96

8. For testing H_0: $\mu = \mu_0$ against $H_1 : \mu < \mu_0$, what is the critical value at $\alpha = 0.01$?

(a) 1.645 (b) –1.645

(c) –2.33 (d) 2.33

9. The hypotheses testing problem H_0: $\mu_0 = 45$ against H_1: $\mu_0 < 45$ be categorized as

(a) left-tailed (b) right-tailed

(c) two-tailed (d) one-tailed

10. When the alternative hypothesis is $H_1 : \mu \neq \mu_0$, the critical region will be determined by

(a) both right and left tails (b) neither right nor left tail

(c) right tail (d) left tail

11. Rejecting H_0, when it is true is called

(a) type I error (b) type II error

(c) sampling error (d) standard error

12. What is the standard error of the sample proportion under H_0?

(a) $\sqrt{\frac{PQ}{n}}$ (b) $\sqrt{\frac{pq}{n}}$

(c) $\frac{PQ}{n}$ (d) $\frac{pq}{n}$

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